EduQuery AI

A Prompt-Based Intelligent Analytics and Prediction Engine for Multi-Campus Academic Databases

Strategic Executive Brief for University Leadership

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Executive Overview

Modern educational networks accumulate extensive operational data across student enrolments, academic examinations, teacher key performance indicators, daily attendance and standardized board outcomes. Institutional decision-makers nevertheless face friction when extracting timely strategic insights because traditional data structures require developer intervention or static reports.

EduQuery AI transforms static database assets into an interactive decision-support engine. It combines a natural-language query interface with machine-learning prediction pipelines so university leadership, Deans and academic administrators can query multi-campus databases using familiar academic terminology and receive analytical summaries, trend visualizations and early-warning student-performance forecasts.

The core engineering stack is open source, but infrastructure and OpenRouter usage are operational costs that require measurement and approval. The proposal's 92.3% predictive figure is an external research benchmark, while EduQuery AI must report its own leakage-safe results. The broader product vision covers 12 institutional analytical domains and prioritizes institutional control of student data.

The August 20 release is now locked as an English-only working MVP. It includes Next.js, FastAPI and Wren AI; semantic-layer documentation; at least 200 evaluated gold prompts; AutoGluon and TabPFN models tracked in MLflow; adaptive model selection; representative SHAP explanations; security evidence; and the final report/live demonstration. The cited 92.3% figure remains an external research benchmark, not an EduQuery AI result.

1. Institutional Context and Strategic Value

1.1 Strategic Alignment and Purpose

EduQuery AI introduces a governed prompt-based analytics layer over existing institutional database infrastructure. It establishes three capabilities:

- **Democratized data access:** Leadership and academic heads can conduct ad-hoc exploration without waiting for custom SQL development.
- **Forward-looking foresight:** Management moves from post-exam review toward earlier intervention using predictive models.
- **Governed metric consistency:** Institutional definitions such as pass percentage and teacher-performance indexes remain consistent across campuses.

1.2 Operational Transformation

Dimension	Traditional workflow	EduQuery AI platform
Query mechanism	Static reports or developer-coded SQL requests	Natural-language prompts grounded in governed schema terms
Turnaround	Days or weeks depending on technical availability	Targeted near-real-time query and chart composition
Analytical horizon	Historical post-facto evaluation	Historical diagnostics plus predictive forecasting
Student support	At-risk identification after final results	Earlier risk detection with explainable indicators
Data sovereignty	Manual spreadsheet exports across silos	Governed execution with database-level access controls

Strategic takeaway: EduQuery AI does not replace institutional databases or academic authority. It provides an intelligent layer that makes existing data more accessible, consistent and actionable.

2. System Architecture and Governed Semantic Layer

2.1 Architecture Overview

Tier	Component	Institutional role
Presentation	Next.js web portal on Vercel	English responsive chat interface, charts and role-scoped dashboards; no protected server-side processing on Vercel
Institutional gateway	FastAPI inside the institution	Authentication, authorization, intent routing, redaction, validation, audit and response composition
Analytics and NL2SQL	Institution-hosted Wren AI plus local RAG	Uses governed MDL, schema glossary, metric definitions and approved examples to produce validated query plans
External model assistance	GPT 5.6 Luna Pro through OpenRouter	Receives only locally de-identified intent/schema context; never student identities, rows, marks, results or predictions
Prediction	Institution-hosted AutoGluon, TabPFN, MLflow and SHAP pipeline	Produces measured performance forecasts, adaptive model-selection evidence and feature attribution
Data and governance	SQL Server `db_aadc11_sqldaanish`	Authoritative academic database with read-only access and row-level controls

2.2 Documents as Source of Truth

The Documents-as-Source-of-Truth framework addresses hallucination and metric inconsistency. The AI does not guess table structures or formulas from generic model knowledge. Prompts are interpreted through a curated semantic model that defines approved tables, relationships, metrics, joins and campus-security parameters before any query reaches the database.

Generated queries must pass schema grounding, allowlisting, Abstract Syntax Tree validation, parameterization, scope enforcement, row limits and read-only database permissions.

3. Institutional Analytical Domains

Domain	Representative question
Student lookup and profile	Retrieve enrolment profile, house assignment and academic status for a roll number
Individual transcript analytics	Display subject marks and historical trajectory across examinations
Class and section aggregates	Generate averages, pass percentages and grade distributions
Cross-campus comparison	Rank schools by subject or class performance
Faculty performance KPIs	Identify sections performing below approved board benchmarks
Board benchmarking	Compare institutional subject results against `BOARD_PER` metrics
Attendance and absenteeism	Identify students whose absence exceeds an approved threshold
Exam operations	Display schedules, deadlines and marks-entry completion status
Audit and data quality	Detect missing marks and incomplete result records
Multi-year trends	Analyse improvement or decline across exams and years
Predictive result forecast	Forecast percentage, grade band and pass probability
Explainable risk attribution	Identify the factors contributing to academic-risk predictions

4. Predictive Intelligence and Explainable AI

4.1 Prediction Engine

The proposed predictive pipeline evaluates AutoGluon Tabular and TabPFN alongside baseline approaches. Candidate models may include XGBoost, LightGBM, CatBoost and neural networks. Final selection must be based on the actual dataset, temporal validation and comparison with a simple historical-average baseline.

4.2 Input Features

- Historical marks, percentages and subject-level trajectories.
- Attendance and absentee patterns.
- Class averages, section size and teacher historical pass rates.
- Difference between campus performance and board benchmarks.

4.3 Explainability

SHAP-based attribution is proposed to show how factors such as attendance, prior performance, teacher context and board gaps influence a forecast. Predictions are advisory decision support only. They must never automatically determine a student's grade, placement or future.

The proposal cites a March 2026 educational-data-mining benchmark reporting 92.32% grade-classification accuracy using AutoGluon. EduQuery AI must publish its own leakage-safe results before making an equivalent product-performance claim.

5. Data Governance, Security and On-Premise Control

5.1 Defence in Depth

Boundary	Control	Outcome
Access control	Institutional authentication, RBAC and campus-scoped authorization	Users access only approved functions and records
Database isolation	Least-privilege read-only SQL identity and SQL Server Row-Level Security where supported	Cross-campus and write access are blocked at the data layer
AI privacy boundary	PII exclusion, approved semantic surface, prompt/result redaction and SQL validation	Sensitive tables and fields are excluded from model context and query execution
Accountability	Tamper-resistant audit events for requests, decisions, generated SQL and outcomes	Administrators can investigate access and failures

5.2 Deployment Sovereignty

The approved MVP uses a split deployment. Vercel hosts the Next.js interface and static assets. FastAPI, Wren AI, local RAG, SQL Server connectivity, model inference, MLflow, SHAP and audit services remain inside the institution. The browser connects through an institution-approved HTTPS/VPN gateway without routing protected prompts through Vercel server functions or analytics.

OpenRouter is an external processor even when zero-data-retention controls are enabled. Therefore local redaction/tokenization runs before every request; the external model receives de-identified schema/intent context only. Student identities, roll numbers, marks, returned rows, predictions and protected prompts never leave the institution. If that exclusion cannot be proven, the external call is blocked.

6. Implementation Roadmap and Team Allocation

6.1 Product Roadmap

Phase	Focus	Institutional deliverable
1	Schema engineering and semantic modelling	Curated analytical views and governed glossary
2	NL2SQL engine and web portal	Working prompt flows for core academic domains
3	Advanced analytics and feature pipelines	Teacher, attendance and board-comparison analytics
4	Prediction and explainability	Validated model, explanation output and model records
5	Security, evaluation and operational readiness	Access-control verification, gold prompts, performance testing and deployment runbook

The August 20, 2026 workforce plan defines the immediate parallel delivery sequence. Multilingual capability, beginning with Urdu, follows after acceptance of the English MVP.

6.2 Team Ownership

Team member	Primary ownership	Secondary support
Muhammad Junaid	NL2SQL engine and governed semantic layer	Schema glossary and SQL validation

Team member	Primary ownership	Secondary support
M. Zain ul Abideen	Prediction pipeline and explainability	Experiment tracking and model evaluation
Abdur Rafay	FastAPI backend and deployment architecture	Authentication, authorization, RLS and audit logging
Abeera Anwer	Next.js portal, visualization and UX	Gold-prompt UX and dashboard evaluation

Codex acts as Workforce and Planner across architecture, task allocation, integration, quality and release readiness. Antigravity executes bounded implementation packets under the relevant human stream owner.

7. Deliverables and Leadership FAQ

7.1 Deliverables

Category	Deliverable
Software	Containerized EduQuery AI web application with Next.js and FastAPI
Semantic governance	Approved schema glossary, metric definitions and relationships
Evaluation	At least 200 versioned gold prompts with expected answers, actual results and measured query accuracy
Prediction	Historical baseline, AutoGluon ensemble and TabPFN artifacts with MLflow tracking, adaptive selection and representative SHAP output
Security	Threat model, RLS verification, access-control tests, SQL-injection results, outbound-payload inspection and PII-exclusion proof
Operations	Deployment configuration, monitoring, backup/recovery and support runbook
Documentation	Technical documentation, stakeholder brief and live demonstration materials

7.2 Leadership FAQ

Does this require proprietary software licensing?
The proposed core stack uses open-source frameworks. Infrastructure, model-provider and operational costs still require measurement and approval.

How is student privacy protected?
Through institution-hosted authorization, a least-privilege read-only database identity, row-level controls, PII exclusion, query validation, redacted logging and fail-closed outbound redaction. Vercel hosts only the interface; OpenRouter receives only de-identified schema/intent context.

What infrastructure is required?
Vercel hosts the interface. Institutional infrastructure runs FastAPI, Wren AI, RAG, SQL connectivity, MLflow and prediction/explainability services. CPU/GPU and storage requirements must be measured against the actual AutoGluon/TabPFN workloads and expected concurrency.

Does the platform replace teachers or administrators?
No. It provides analytics and predictive decision support. Human educators retain complete academic authority.